

Hall FOC (MPOS) User Guide

Power Application Controller™





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1 DESCRIPTION

The Hall FOC software module is one of the selectable MPOS (Motor Position) options available to the user for motor control using the Active-Semi FOC firmware. When selected, the Hall sensor inputs read the signals from Hall sensors fitted inside the motor assembly.

This method of sensing the rotor position is typically used in applications where:

- 1) The motor must operate at speeds below the capability of sensor-less motor position methods
- 2) Align&Go startup method is not an option because of very dynamic startup load or backward movement of the rotor during align is not tolerated

The Hall mode automatically implements angle interpolation in order to minimize the phase current distortion that results on every hall state transition.

The Hall FOC module can be used to drive the motor in 2 modes of operation:

- 1) Throughout the entire speed range
- 2) Together with the sensorless Estimator where the mode is switched dynamically so that most of the speed range is covered by the Estimator mode except for low speeds that are below the estimator range which are covered by the Hall mode.



2 FIRMWARE IMPLEMENTATION

2.1 Hall FOC Module

The FOC firmware Hall sensor module processes the information from the hall sensor inputs to calculate the motor position and speed as outlined in the following steps:

The first control cycle after a hall state transition occurs:

- Check for hall state sequence or invalid state faults
- Calculate the electrical angle depending on the hall state and hall delay compensation.
- Angle correction is applied with a constant offset and a speed dependent component.

The second control cycle after the hall state transition occurs:

- Rotor speed is calculated from the time measured between hall state transitions and a low-pass filter is applied.
- Angle interpolation is applied using the calculated motor speed

All subsequent control cycles until the next hall state transition occurs:

- Angle interpolation is applied using the calculated motor speed

A block diagram that describes the basic flow of the hall process function is shown in Figure 1.

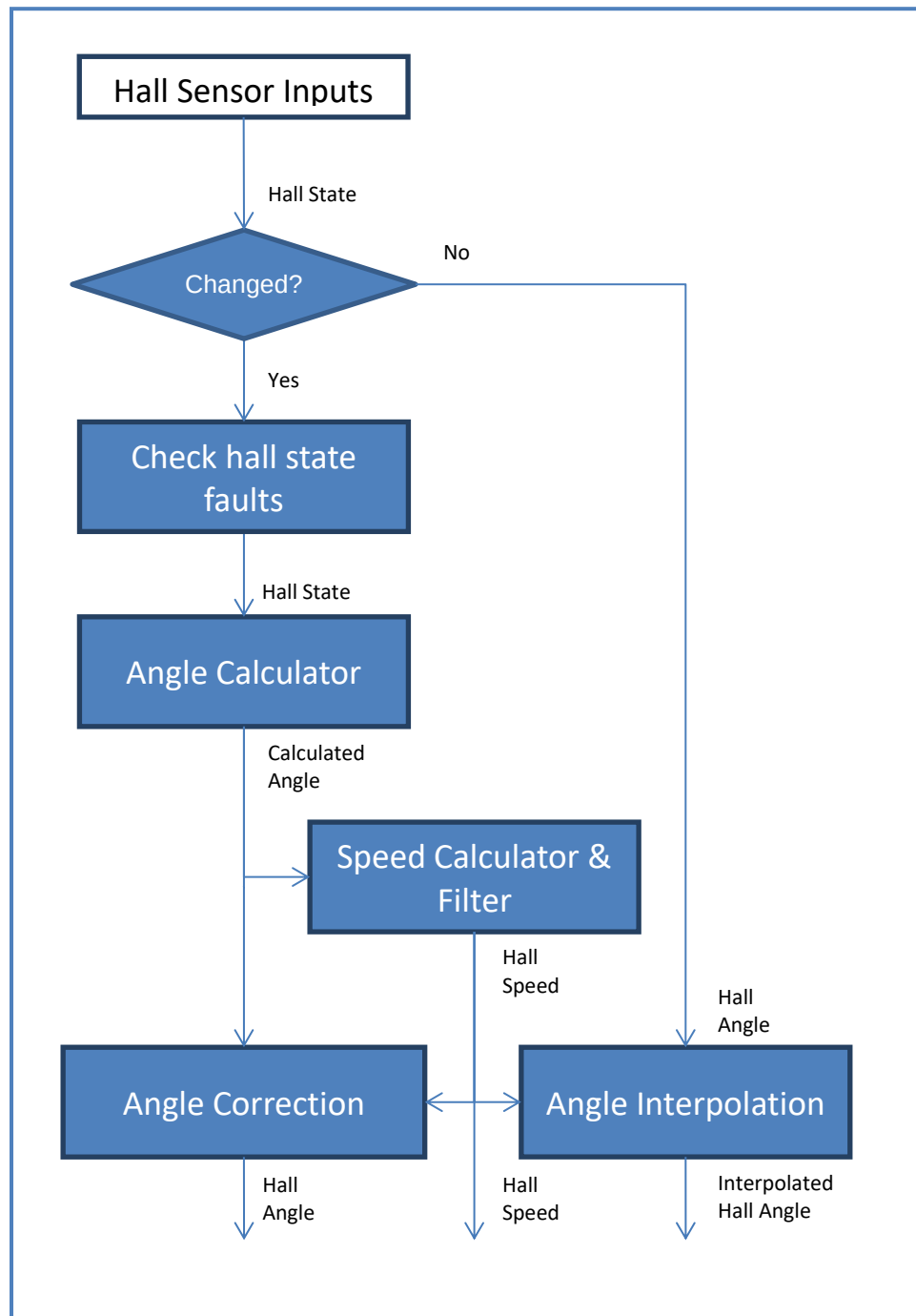


Figure 1: Hall Process Function Diagram

2.2 Startup Sequence

When first starting the motor, the hall sensor inputs are used to determine the 60-degree sector that the rotor is in. This is used to start driving the motor at the correct sector in torque mode. Initially the user-configurable starting current target is used. After the also user-configurable number of hall transitions have occurred, the control is switched to normal torque or speed mode.

Figure 2 depicts the sequence followed during the motor startup process from zero-speed to Hall mode.

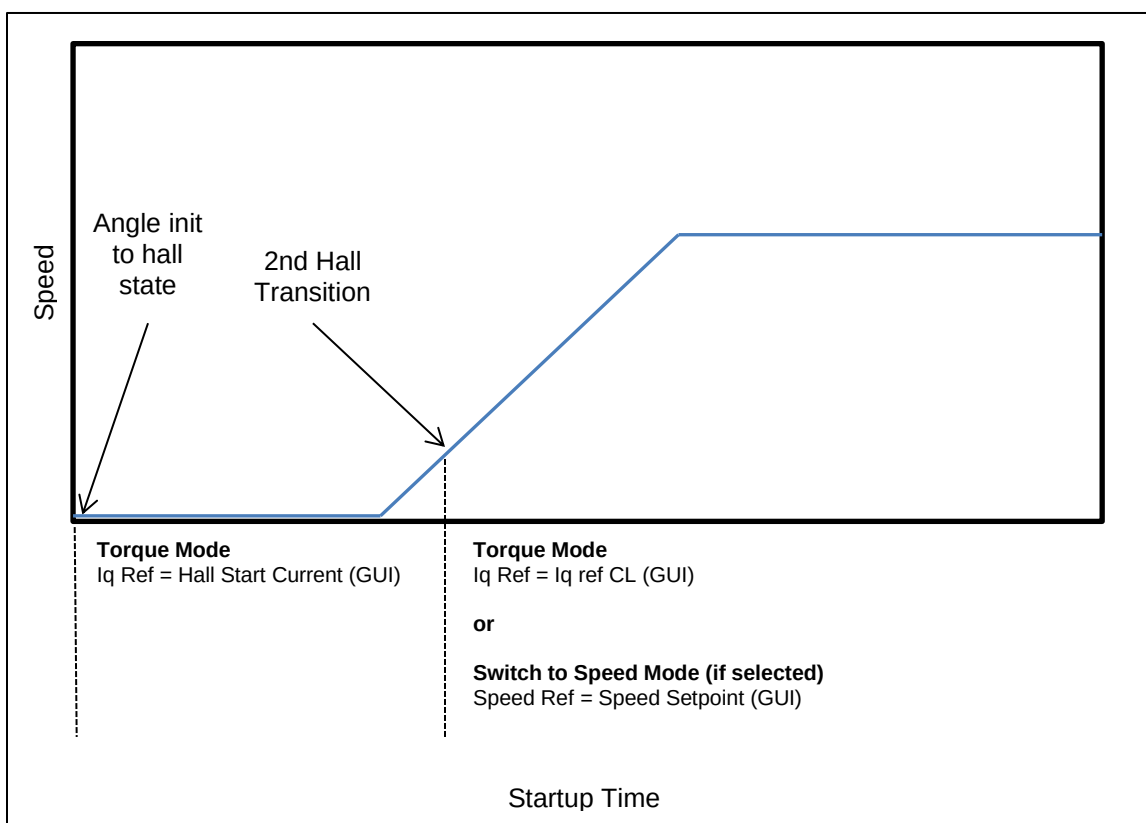


Figure 2: Motor Startup Sequence in Hall Mode



2.3 Inputs

The Hall process block receives the following inputs during runtime:

Input Variable Name	Format	Units
SensorStateNew	Q0	-

2.4 Outputs

The Hall process block produces the following outputs:

Output Variable Name	Format	Units
hall_angle	Q16 (1.0 = 65536)	rad $[-\pi, \pi]$
hall_spd_radps	Q16 (1.0 = 65536)	rad/sec
hall_spd_scaled_q14	Q14 (1.0 = 16384)	Normalized speed $[-1, 1]$
hall_spd_scaled_q28	Q28 (1.0 = 268435456)	Normalized speed $[-1, 1]$

2.5 Faults

2.5.1 Invalid State Fault

Invalid state faults can be generated by two different checks:

- 1) Invalid hall state
Valid hall states are defined as any value between 1 and 6. If a state is read as 0 or 7, this is considered to be an invalid state and the motor will be disabled.
- 2) Invalid sequence fault
When the motor is running, the hall states must follow a sequence. The sequence check verifies whenever a state transition is detected that the new state is following the sequence. If an invalid state sequence is detected, the motor is disabled. Note that the sequence check verifies that the new state is either the expected next state or previous state. This ensures that the fault will be falsely triggered in the case that the motor moves backward even when the direction is set to forward.

3 HARDWARE CONFIGURATION

3.1.1 Timer Block

The Hall Timer block is used to measure the time between hall state transitions which in turn is used to calculate the motor speed. The main steps in configuring the Hall Timer block are summarized below:

- 1) Configure timer clock source and divider. The divider typically needs to be large enough to allow measurement of the slowest motor speed without overflowing.
- 2) Set timer to UP mode
- 3) Enable the timer base interrupt to indicate timer overflow at low speeds
- 4) Enable NVIC interrupt for hall timer and configure priority. The priority for this interrupt can be same as the application level loop.

If the timer overflows and triggers the base interrupt, the firmware considers this to be essentially 0 speed since it is below the speed that it can detect. In this case the speed is set to 0 and the angle interpolation is reset.

The hall timer configuration is done in “hall_timer_config.h”.

3.1.2 I/O Ports

Hall sensor operation requires the use of 3 port pins:

- 3 inputs for hall signals A ,B, and C

All 3 port pins used for hall operation are configured as follows:

- 1) Configure the 3 port pins as GPIO inputs.
- 2) Enable internal pullups for the 3 port pins.

The I/O port configuration is done in “hall_io_config.h”

4 FIRMWARE CONFIGURATION

4.1 Selecting the MPOS Hall Mode

The Hall module is enabled in “config_app.h” by selecting one of the following MPOS mode definitions depending on the behavior desired:

- 1) **#define** MPOS_SELECT MPOS_HALL_ONLY

This MPOS option runs only in interpolated hall sensor mode throughout the entire speed range.

- 2) **#define** MPOS_SELECT MPOS_LBG1_START_HALL

This MPOS option runs in interpolated hall sensor mode to start the motor and switches to sensorless mode when the speed is above a user-configurable threshold.

4.2 Compile Options and Parameters

The following compile options are available in “config_app.h” to configure the hall module depending on the hall MPOS mode selected:

- 1) Hall Only Mode & Hall + Estimator Mode

The options below are available for both Hall Only and Hall + Estimator MPOS Modes:

ENABLE_ANGLE_CORRECTION

Apply angle correction to calculated angle consisting of a constant offset component and a speed-proportional component.

ENABLE_HALL_SEQUENCE_CHECK

Verifies that the next hall state detected follows the expected hall sequence. If the check fails, the motor is disabled.

HALL_METHOD_SELECT

The detection method for hall state transitions can be configured by selecting one of the following hall method definitions:

- 1) **#define** HALL_METHOD_SELECT HALL_METHOD_INTERRUPT
Uses GPIO interrupts to detect when there is a transition in any of the hall sensor signals. This is the recommended method as it significantly reduces phase current distortion at high speed.

- 2) **#define** HALL_METHOD_SELECT HALL_METHOD_POLLING
Uses polling of the hall sensor signals to detect the hall state at the start of each control cycle.

ENABLE_HALL_ANGLE_DELAY_COMPENSATION

Add delay compensation to hall angle. This option is highly recommended especially necessary in high speed applications to reduce phase current distortion due to angle delay error. The type of delay compensation applied depends on the hall method selected:



1) HALL_METHOD_INTERRUPT

Compensates for control loop execution delay time and control loop start time relative to hall edge.

2) HALL_METHOD_POLLING

Compensates for control loop execution delay time only.

CONFIG_HALL_TIMEOUT_EFREQ_HZ

Configures the lowest speed that the hall timer is able to measure before declaring that the motor is not moving.

CONFIG_HALL_TIMEOUT_MDISABLE

If enabled, configures the hall engine to disable the motor if the hall timer reaches timeout.

2) Hall + Estimator Mode

The options below are additional options available when Hall + Estimator Mode is selected:

SWITCHOVER_SPD_HALLEST_HYST_PCT_Q16

Hysteresis band for switching from Estimator to Hall mode as a percentage of user-defined switchover speed.

ENABLE_ANGLEDIFF_SPEED_CHECK

Check that the angle and speed are within an error threshold before switching between estimator and hall modes. If the check fails, warning is reported. When enabled, the following options can be configured:

ANGLEDIFF_SPEED_ERROR_PCT_Q16

Allowed speed error band as a percentage of the switchover speed and entered in decimal format. Recommended value = 0.1.

ANGLEDIFF_ANGLE_ERROR_DEG_Q12

Allowed angle error band in radians. Recommended value = $\pi/10$.

ENABLE_OLSPEED_ANGLEDIFF_TIMEOUT

Enables timeout for anglediff and speed check. If the check fails for the specified timeout period, the motor is disabled. When enabled, the following option can be configured:

ANGLEDIFF_TIMEOUT_SEC

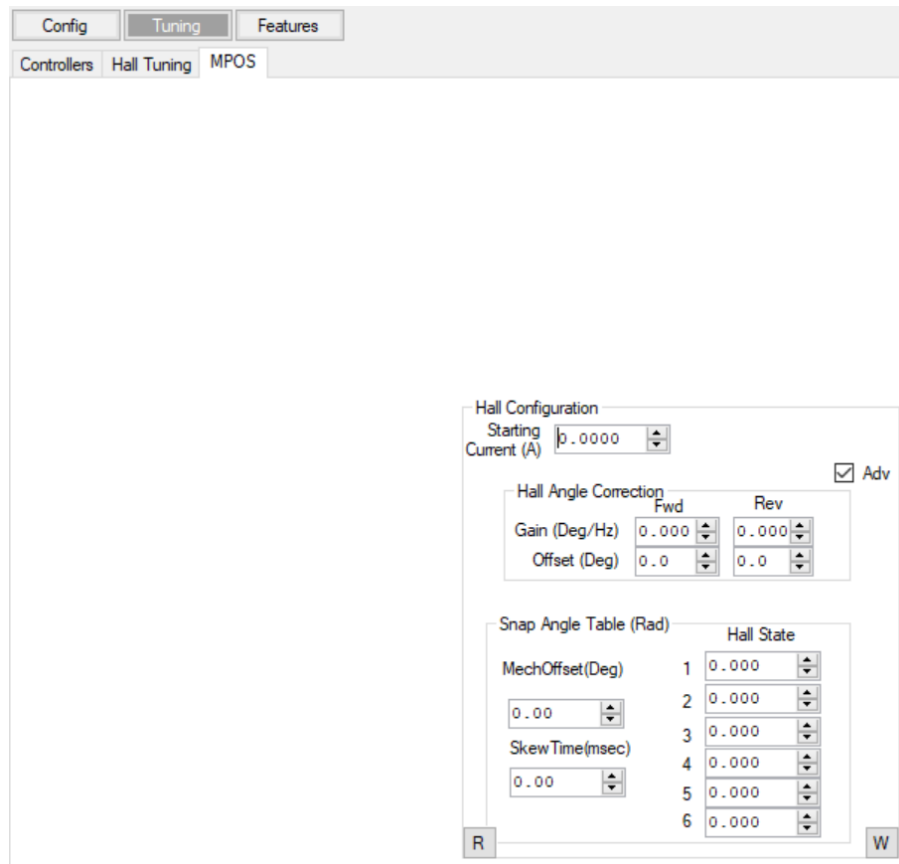
Timeout period defined in seconds. Recommended value = 2.

4.3 GUI Parameters

The GUI contains a section where the user can configure the parameters for Hall operation which can vary by motor. The Hall configuration section in the GUI can be found by selecting the “Tuning” view and then the “Tuning Config” tab. The parameters available for configuration in the Hall section depend on the MPOS method selected.

4.3.1 Hall Parameters for Hall Only Mode

The Hall configuration view for Hall Only mode is shown in *Figure 3* and the configurable parameters are listed below:



Hall Configuration

Starting Current (A) 0.0000

☒ Adv

Hall Angle Correction

	Fwd	Rev
Gain (Deg/Hz)	0.000	0.000
Offset (Deg)	0.0	0.0

Snap Angle Table (Rad)

MechOffset (Deg)	Hall State
0.00	1 0.000
	2 0.000
	3 0.000
	4 0.000
	5 0.000
	6 0.000

SkewTime (msec) 0.00

R W

Figure 3: Hall Only Configuration Section in GUI

Starting Current (A)

Current applied to motor phase during first startup cycles to ensure that the rotor is gets unlocked into motion.

Angle Correction Gain Fwd/Rev (Deg/Hz)

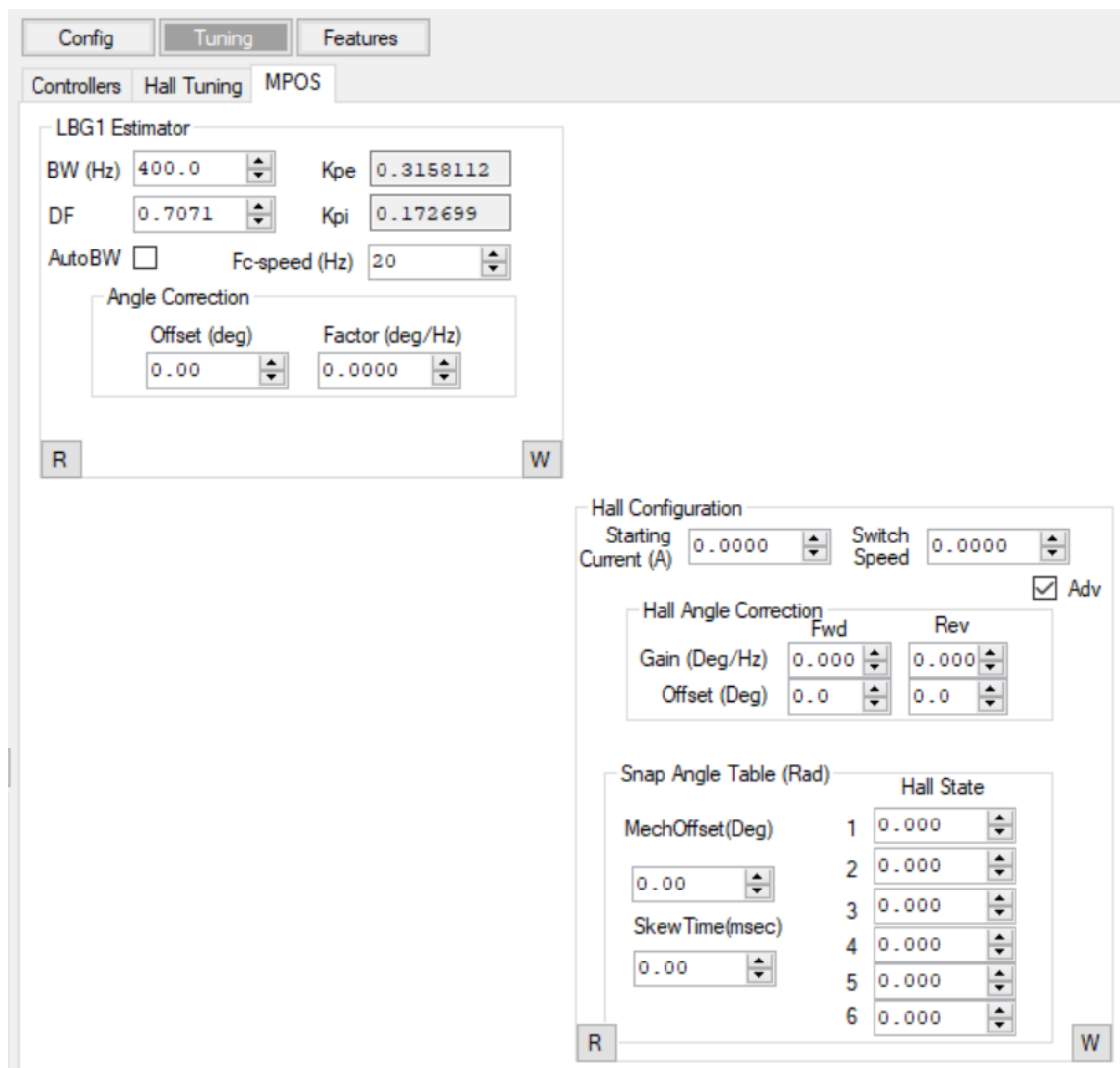
This is the speed-dependent offset gain applied to the hall angle in the forward and reverse directions respectively. The user will need to tune this value to optimize the calculated angle.

Angle Correction Offset Fwd/Rev (Deg)

This is the fixed offset applied to the hall angle in the forward and reverse directions respectively. The user will need to tune this value to optimize the calculated angle.

4.3.2 Hall Parameters for Hall + Estimator Mode

The Hall configuration view for Hall + Estimator mode is shown in Figure 4. It is very similar to the Hall Only configuration view except that some additional configuration items become available:



The screenshot displays the 'Hall + Estimator Configuration' section of the GUI. It is divided into two main panels: 'LBG1 Estimator' and 'Hall Configuration'.

LBG1 Estimator Panel:

- Controllers:** Config, Tuning, Features
- Sub-panels:** Controllers, Hall Tuning, MPOS
- Parameters:**
 - BW (Hz): 400.0
 - DF: 0.7071
 - Kpe: 0.3158112
 - Kpi: 0.172699
 - AutoBW: ☐
 - Fc-speed (Hz): 20
 - Angle Correction:**
 - Offset (deg): 0.00
 - Factor (deg/Hz): 0.0000
- Buttons:** R, W

Hall Configuration Panel:

- Parameters:**
 - Starting Current (A): 0.0000
 - Switch Speed: 0.0000
 - ☒ Adv
 - Hall Angle Correction:**
 - Fwd: Gain (Deg/Hz) 0.000, Offset (Deg) 0.0
 - Rev: Gain (Deg/Hz) 0.000, Offset (Deg) 0.0
 - Snap Angle Table (Rad):**
 - MechOffset(Deg): 0.00
 - SkewTime(msec): 0.00
 - Hall State:**

1	0.000
2	0.000
3	0.000
4	0.000
5	0.000
6	0.000
- Buttons:** R, W

Figure 4: Hall + Estimator Configuration Section in GUI

Switch Speed (Hz)

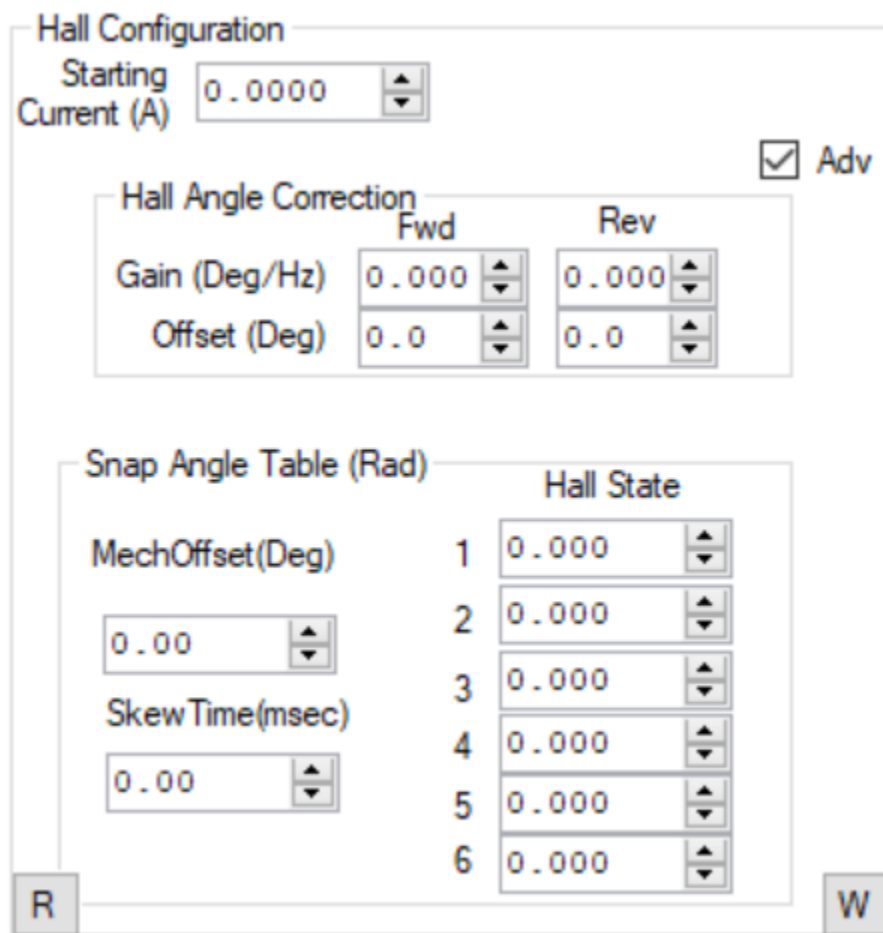
This is the speed at which switchover from Hall Mode to Estimator Mode is desired to occur.

Estimator Configuration Group

This group of configuration parameters needs to be tuned according to the User Guide for the Estimator.

4.3.3 Advanced View

The advanced panel of the Hall configuration view for both Hall Only and Hall + Estimator mode is shown in Figure 5. This view can be used by the user if it is necessary to independently and manually adjust the hall angle table.



Hall Configuration

Starting Current (A)

☒ Adv

Hall Angle Correction

	Fwd	Rev
Gain (Deg/Hz)	0.000	0.000
Offset (Deg)	0.0	0.0

Snap Angle Table (Rad)

MechOffset(Deg)	SkewTime(msec)	Hall State
0.00	0.00	1 0.000
		2 0.000
		3 0.000
		4 0.000
		5 0.000
		6 0.000

R W

Figure 5: Advanced Panel in Hall Configuration

5 TUNING

5.1 Description

During the hall tuning procedure, the motor is turned in open loop mode using the Align&Go startup method. While the motor is turning in open loop, the hall signal transitions are detected and used to automatically define the following for the motor being tuned:

- 1) Hall angle table
- 2) Hall state sequence

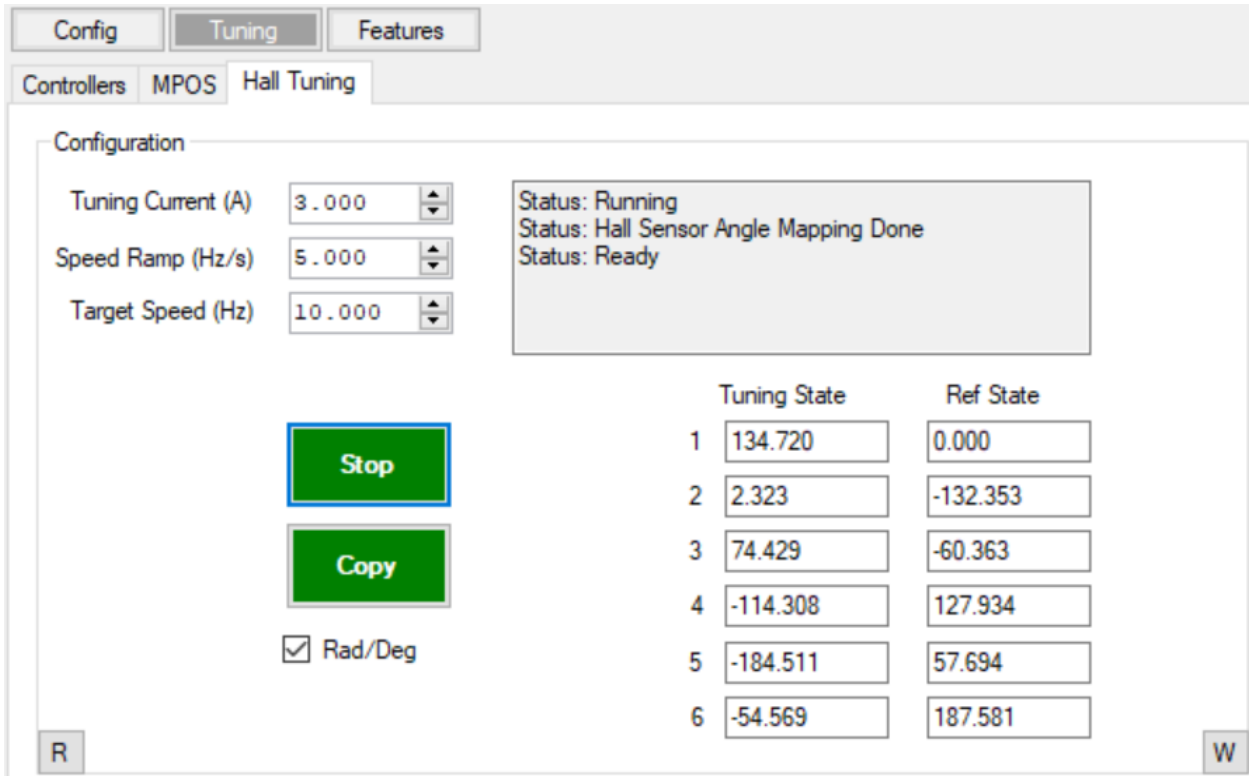
5.2 Enabling the Hall Tuning Tool

The hall tuning tool can be enabled by making sure that the following definition is enabled in "config_features.h":

```
#define ENABLE_HALL_TUNING
```

5.3 GUI Parameters

When the hall tuning tool is enabled the tab shown in Figure 6 will become available in the GUI.



	Tuning State	Ref State
1	134.720	0.000
2	2.323	-132.353
3	74.429	-60.363
4	-114.308	127.934
5	-184.511	57.694
6	-54.569	187.581

Figure 6: Hall Tuning Tab

5.4 Tuning Procedure

In order to run the hall tuning tool, please follow the following steps:

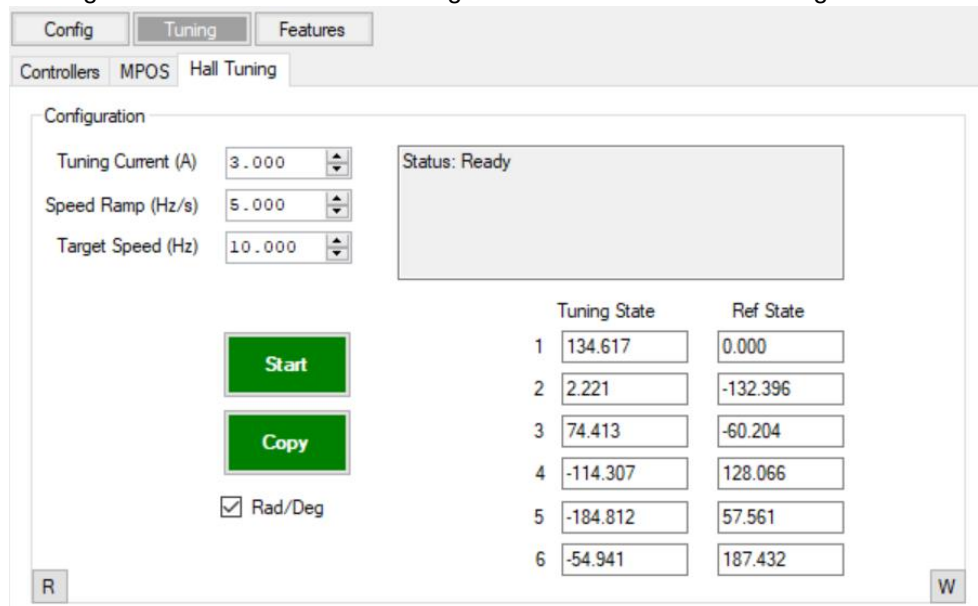
- 1) Enter a value for the configuration parameters:
 - a. **Tuning Current (A)** : This is the current applied to the motor phases while turning in open loop to detect the hall signal transitions. A good starting value for this parameter is about 20% of maximum current. If the motor stalls while running in open loop, this value may need to be increased.
 - b. **Speed Ramp (Hz/s)** : This is the desired acceleration rate during the open loop speed ramp to target. A good way to choose this value may be such that the time to ramp from 0 to the target speed is about 5 seconds.
 - c. **Target Speed (Hz)** : This is the desired motor speed to run while detecting the hall signal transitions. An appropriate value may be about 10% of the top motor speed.
 - d. **Rad/Deg display selection**: This enables the degree displaying in the unit of Rad or Degree.
- 2) Click the "Start" button. The motor will start running in open loop mode and ramping to the target speed.
- 3) As the hall angles start getting detected, angle values in radians will start showing up in the Hall State and Snap State displays. The Hall State angles are the open loop forced angles at which the corresponding hall state transition is getting detected. These angles are then "snapped" to the next angle value in the internal reference hall angle table resulting in the Snap State angles. While the hall angles are being processed the status Hall Tuning status window will display "Detecting Hall Sensor Angles".
- 4) When the process is complete the Hall Tuning status window will display "Hall Sensor Angle Mapping Done". At this point the process can be stopped by clicking the "Stop" button.
- 5) In order to copy the snap angle table into the GUI for running in hall mode click the "Copy" button.
- 6) The motor only needs to be run in **forward mode only** for HALL tuning since the same Hall angle table can be used for the reverse mode.
- 7) Start the motor to a slow speed in torque mode. At this slow speed try adjusting the Angle Correction Offset to verify that the hall angles have been detected properly. As the offset is adjusted, if the motor starts turning faster, it means that the angle is shifting to a more correct position. When the fastest possible speed is achieved with a given Iq Reference, this is the desired offset for this configuration. There are 3 possible reasons why the offset may need to be adjusted:
 - a. It is possible that depending on the chosen values for Tuning Current and on the Target Speed that the hall angles may have been "snapped" incorrectly to the previous or to the next sector, in which case the motor will not be running efficiently with the default values. If this is the case the final offset value may end up being close to either -60 or + 60 degrees.
 - b. The hall sensors are not accurately placed inside the motor resulting in the need to shift the angle by possibly a few degrees in either positive or negative direction.
 - c. There are some motors where the hall sensor is shifted intentionally by 30 degrees in which case the resulting optimal offset may be either -30 or +30 degrees.

5.5 Tuning Example (48V E-bike motor)

To simplify the tuning (skipping MID stage), the .xml from FOC v4.3 could be used for the new FOC firmware. Otherwise, please follow the general guidelines on how to get motor parameters with the auto-tuning guide in “..\resources\guides\features”.

During the hall tuning, there are two main results to capture: HALL offset angle while detecting Phase U HALL rising edge and the angle mapping from HALL states. Hence, here are the procedures to get these two results:

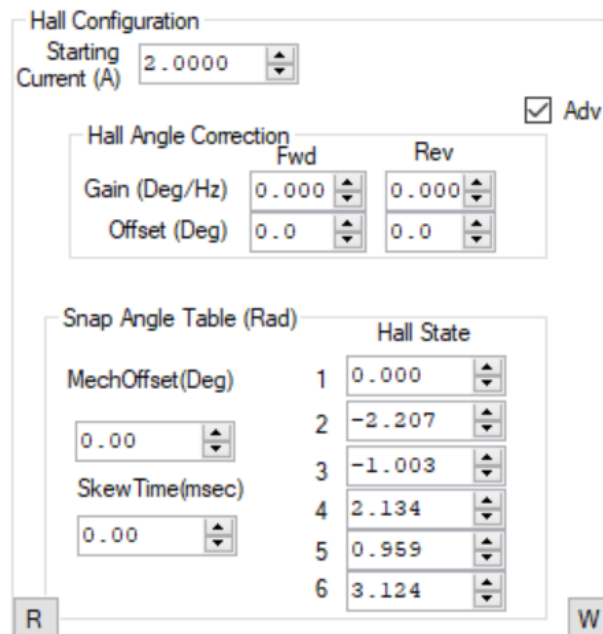
1. Load the .xml file and Write All Parameters. It is recommended to double check if the parameters are looking good after writing.
2. To get the HALL offset angle, please run the motor without Load as below (Figure 5.5.1):
 - A. Configure Tuning Current (A) as 3A.
 - B. Target Speed as 10Hz (Slow RPM speed).
 - C. Click the “START” button on the Hall Tuning Tab.
 - D. Monitor if the motor is rotating smoothly. If not, please gradually increase Target speed (i.e, 10Hz or 20Hz) until the motor is rotating smoothly.
 - E. The HALL offset angle is displayed in Box 1 of Tuning State as highlighted in Yellow in Figure 5.5.1. For instance, 134.617 here is with unit of Degree If Rad/Deg checkup is selected.
 - F. Please type the above HALL offset angle into Box of **MechOffset(Deg)** on Hall Configuration Tab as illustrated in Figure 5.5.2. This is the offset angle for HALL sensor.



The screenshot shows the 'Hall Tuning' tab in the software interface. It includes configuration fields for Tuning Current (A), Speed Ramp (Hz/s), and Target Speed (Hz). A 'Status: Ready' box is present. Below the configuration fields are 'Start' and 'Copy' buttons, and a checked 'Rad/Deg' checkbox. To the right, a table displays the 'Tuning State' and 'Ref State' for six HALL sensors.

	Tuning State	Ref State
1	134.617	0.000
2	2.221	-132.396
3	74.413	-60.204
4	-114.307	128.066
5	-184.812	57.561
6	-54.941	187.432

Figure 5.5.1 Hall angle offset tuning



Hall Configuration

Starting Current (A)

☒ Adv

Hall Angle Correction

	Fwd	Rev
Gain (Deg/Hz)	0.000	0.000
Offset (Deg)	0.0	0.0

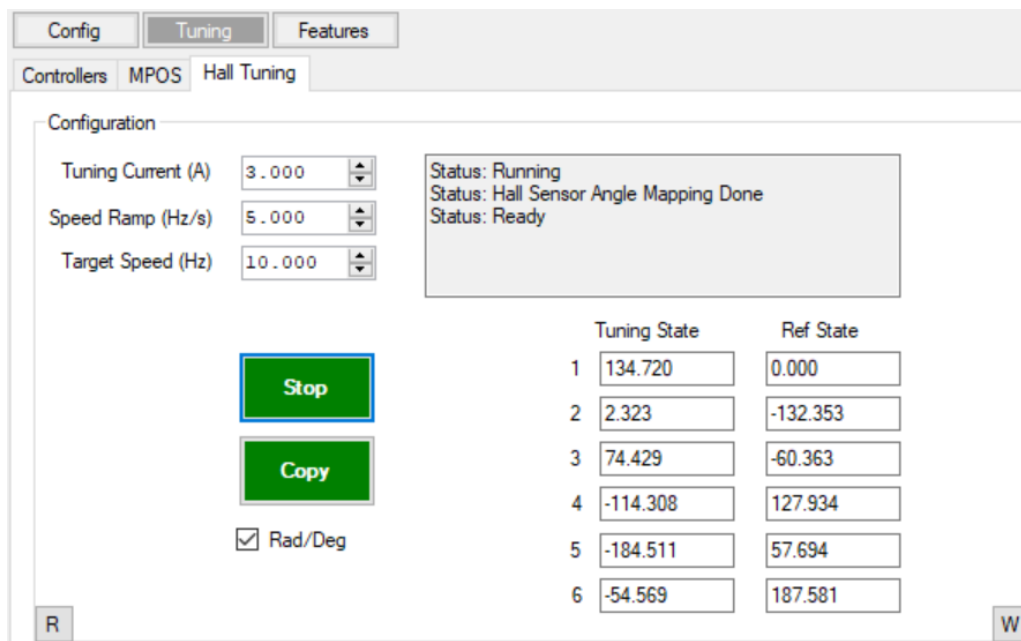
Snap Angle Table (Rad)

MechOffset(Deg)	Hall State
0.00	1 0.000
0.00	2 -2.207
0.00	3 -1.003
0.00	4 2.134
0.00	5 0.959
0.00	6 3.124

R W

Figure 5.5.2 HALL offset angle for Hall operation

3. To tune the angle mapping from six HALL states, we need to run the motor with high speed reference as follow:
 - A. Configure Target Speed as 40Hz or higher.
 - B. Be sure the motor is rotating smoothly.
 - C. Once the motor speed is stabled, please monitor the Ref state in Figure 5.5.3. These six states are the mapping angles in Rad or Degree for each Hall state transition.
 - D. Click the "STOP" button and then Click "Copy" button. The Snap state will be copied for HALL operation as illustrated in Figure 5.5.4.



Config **Tuning** **Features**

Controllers **MPOS** **Hall Tuning**

Configuration

Tuning Current (A)

Speed Ramp (Hz/s)

Target Speed (Hz)

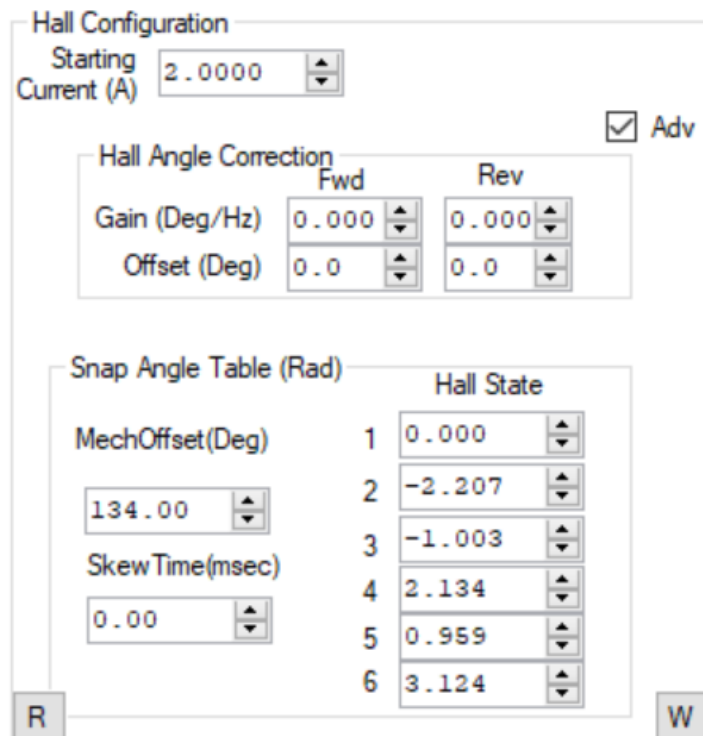
Status: Running
Status: Hall Sensor Angle Mapping Done
Status: Ready

☒ Rad/Deg

	Tuning State	Ref State
1	134.720	0.000
2	2.323	-132.353
3	74.429	-60.363
4	-114.308	127.934
5	-184.511	57.694
6	-54.569	187.581

R W

Figure 5.5.3 Hall angle mapping with six HALL states



The screenshot shows the 'Hall Configuration' window with the following settings:

- Starting Current (A):** 2.0000
- Hall Angle Correction:**
 - Gain (Deg/Hz):** Fwd: 0.000, Rev: 0.000
 - Offset (Deg):** Fwd: 0.0, Rev: 0.0
- Snap Angle Table (Rad):**

MechOffset(Deg)	Hall State
134.00	1: 0.000
	2: -2.207
	3: -1.003
	4: 2.134
	5: 0.959
	6: 3.124
- SkewTime(msec):** 0.00

Buttons 'R' and 'W' are at the bottom left and right respectively. A checkbox 'Adv' is checked in the top right.

Figure 5.5.4 Hall mapping angle copied from the HALL tuning tab (Please check “Adv” for display the detail)

HALL offset angle from the Step 2 may not be accurate because of the motor’s cogging torque and HALL signal skewing during the slow speed region. Hence, it is recommended to optimize the HALL offset angle further:

- Set the **Starting Current** as 2A or higher as illustrated in Figure 5.5.4.
- With the above hall tuning results, the motor can be operated in Torque mode with **Iq ref CL** as 1A and **Id ref CL** as 0A. Once “Saturation warning” is displayed and running smoothly, then move on to next step.
- Stop the motor.
- Set **Iq ref CL** with a small reference (such as 0.7A or 0.8A) and run the motor again. Be sure the motor is rotating without the saturation warning.
- Adjust Box **MechOffset (Deg)** in Hall Configuration Tab (Figure 5.5.4) gradually to find the best angle, which can provide the highest speed. (For example, the optimized offset angle could be +-20 degrees from the Hall tuning result)
- Skew Time (msec) in Hall Configuration Tab is default as zero by assuming the Hall state is changed instantaneously.



CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

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